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EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS
Las Vegas

Fast Focus: Caching Options in .NET

Jason Bock

Staff Software Engineer
Rocket Mortgage

Level: Intermediate

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Personal Info

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Downloads

https://github.com/JasonBock/CachingOptions

Overview

- Goal
- Demo

Remember…
https://github.com/JasonBock/CachingOptions

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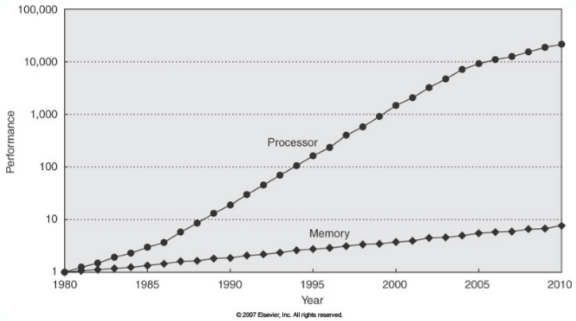
Caching Options in .NET

GOAL

20
40
60
80
100
120
140
160
180
200
220
240
km/h

CPU, Cache, Memory



L1i/d$: 32KB, L2: 256KB, L3: 8MB*

Latency numbers every programmer should know

```
L1 cache reference ......................... 0.5 ns
Branch mispredict ............................ 5 ns
L2 cache reference ........................... 7 ns
Mutex lock/unlock ........................... 25 ns
Main memory reference ...................... 100 ns
Compress 1K bytes with Zippy ............. 3,000 ns  =   3 µs
Send 2K bytes over 1 Gbps network ....... 20,000 ns  =  20 µs
SSD random read ........................ 150,000 ns  = 150 µs
Read 1 MB sequentially from memory ..... 250,000 ns  = 250 µs
Round trip within same datacenter ...... 500,000 ns  = 0.5 ms
Read 1 MB sequentially from SSD* ..... 1,000,000 ns  =   1 ms
Disk seek ........................... 10,000,000 ns  =  10 ms
Read 1 MB sequentially from disk .... 20,000,000 ns  =  20 ms
Send packet CA->Netherlands->CA .... 150,000,000 ns  = 150 ms
```

Summary: Instructions matter; memory matters more
(and I/O dwarfs them all…)

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DEMO: CACHING STRATEGIES

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